thinking, "That's a pretty big territory; does the field of social psychology really cover *all* this?" As you will soon see, it does, so we are not exaggerating: social psychology truly does investigate the entire span of social existence—a true rainbow of human social experience—but with the individual as the focus.

At this point, we hope we have whetted your appetite for the "moveable feast" that will follow, so we'd like to plunge right in and begin addressing topics and questions like the ones mentioned above. Before doing so, though, we feel it's important to provide you with some background information about the scope, nature, and methods of our field. This information will be useful to you in reading the entire book (as well as in your course), and in understanding how social psychologists go about answering fascinating questions about the social side of life, so it is crucial that we provide it here. To be efficient and hold these tasks to a minimum, we'll proceed as follows.

First, we present a more formal definition of social psychology—what it is and what it seeks to accomplish. Second, we'll describe several current trends in social psychology. These are reflected throughout this book, so knowing about them at the start will help you recognize them and understand why they are important. Third, we examine some of the methods used by social psychologists to answer questions about the social side of life. A working knowledge of these basic methods will help you to understand how social psychologists add to our understanding of social thought and social behavior, and will also be useful to you outside the context of this course. Then, we provide you with an overview of some of the special features in this book—features we think you will find helpful in many ways.

Social Psychology: An Overview

Providing a definition of almost any field is a complex task. In the case of social psychology, this difficulty is increased by two factors: the field's broad scope and its rapid rate of change. As you will see in every chapter of this book, social psychologists truly have a wide range of interests. Yet, despite this fact, most focus mainly on the following task: understanding how and why individuals behave, think, and feel as they do in social situations—ones involving the actual presence of other people, or their symbolic presence. Accordingly, we define social psychology *as the scientific field that seeks to understand the nature and causes of individual behavior, feelings, and thought in social situations*. Another way to put this is to say that *social psychology investigates the ways in which our thoughts, feelings, and actions are influenced by the social environments in which we live—by other people or our thoughts about them* (e.g., we imagine how they would react to actions we might perform). We'll now clarify this definition by taking a closer look at several of its key aspects.

Social Psychology Is Scientific in Nature

What is *science?* Many people seem to believe that this term refers only to fields such as chemistry, physics, and biology—ones that use the kind of equipment shown in Figure 1.2. If you share that view, you may find our suggestion that social psychology is a scientific discipline somewhat puzzling. How can a field that seeks to study the nature of love, the causes of aggression, and everything in between be scientific in the same sense as chemistry, physics, or computer science? The answer is surprisingly simple.

In reality, the term *science* does not refer to a special group of highly advanced fields. Rather, it refers to two things: (1) a set of values and (2) several methods that can be used to study a wide range of topics. In deciding whether a given field is or is not scientific, therefore, the critical question is, Does it adopt these values and methods? To the extent it does, it is scientific in nature. To the extent it does not, it falls outside the realm of science. We examine the procedures used by social psychologists in their research in detail in a later section, so here we focus on the core values that all fields must adopt to





FIGURE 1.2 What Is Science, Really?
Many people seem to believe that only fields that use sophisticated equipment like that shown (left) can be viewed as scientific. In fact, though, the term science *simply refers to adherence to a set of basic values (e.g., accuracy, objectivity) and use of a set of basic methods that can be applied to almost any aspect of the world around us—including the social side of life. In contrast, fields that are not scientific in nature (right) do not accept these values or use these methods.*

be considered scientific in nature. Four of these are most important:

Accuracy: A commitment to gathering and evaluating information about the world (including social behavior and thought) in as careful, precise, and error-free a manner as possible.

Objectivity: A commitment to obtaining and evaluating such information in a manner that is as free from bias as humanly possible.

Skepticism: A commitment to accepting findings as accurate only to the extent they have been verified over and over again.

Open-mindedness: A commitment to changing one's views—even views that are strongly held—if existing evidence suggests that these views are inaccurate.

Social psychology, as a field, is deeply committed to these values and applies them in its efforts to understand the nature of social behavior and social thought. For this reason, it makes sense to describe it as scientific in orientation. In contrast, fields that are not scientific make assertions about the world, and about people, that are not put to the careful test and analysis required by the values listed above. In such fields—ones like astrology and aromatherapy—intuition, faith, and unobservable forces are considered to be sufficient (see Figure 1.2) for reaching conclusions—the opposite of what is true in social psychology.

"But why adopt the scientific approach? Isn't social psychology just common sense?" Having taught for many years, we can almost hear you asking this question. And we understand why you might feel this way; after all, each of us has spent our entire lives interacting with other people and thinking about them, so in a sense, we are all amateur social psychologists. So, why don't we just rely on our own experience and intuition as a basis for understanding the social side of life? Our answer is straightforward: Because such sources provide an inconsistent and unreliable guide to understanding social behavior and social thought. Why? In part because our own experiences are unique and may not provide a solid foundation for answering general questions such as "Why do we sometimes go along 'with the group' even if we disagree with what it is doing?" "How can we know what other people are thinking or feeling at any given time?" In addition, common sense often provides inconsistent and contradictory ideas about various aspects of social life. For instance, consider the statement "Absence makes the heart grow fonder." Do you agree? Is it true that when people are separated from those they love, they miss them and so experience increased longing for them? Many people would agree. They would answer "Yes, that's right. Let me tell you about the time I was separated from…" But now consider the statement "Out of sight, out of mind." How about this one? Is it true? When people are separated from those they love, do they quickly find another romantic interest? (Many popular songs suggest that this so—for instance, in the song "Love the One You're With" written and recorded by Stephen Stills, he suggests that if you can't be with the person you love, you should love the person you are with.) As you can see, these two views—both suggested by common sense and popular culture—are contradictory. The same is true for many other informal observations about human behavior—they seem

plausible, but often the opposite conclusion seems equally possible. How about these: "Two heads are better than one" and "Too many cooks spoil the broth." One suggests that when people work together, they perform better (e.g., make better decisions). The other suggests that when they work together, they may get in each other's way so that performance is actually reduced. Here's one more: Is it "Familiarity breeds content" (as we come to know others better, we tend to like them more—we feel more comfortable with them), or is it "Familiarity breeds contempt" (as we come to know others better, we tend to like them less). Common sense suggests that "more is more" where liking is concerned—the more familiar we are with others, the more we tend to like them, and there is some support for this view (see Chapter 7). On the other hand, though, research findings indicate that sometimes, the more we know about others (the better we come to know them), the less we like them (Norton, Frost, & Ariely, 2006). Why? Because as we learn more about others we recognize more ways in which we are dissimilar to them, and this growing awareness of dissimilarity causes us to notice yet more ways in which we are dissimilar, which leads to disliking.

We could continue, but by now, the main point should be clear: Common sense often suggests a confusing and inconsistent picture of human behavior. This doesn't mean that it is necessarily wrong; in fact, it often does offer intriguing clues and insights. But it doesn't tell us when various principles or generalizations hold—when, for instance, "Absence makes the heart grow fonder" and when it leads to "Out of sight, out of mind." Only a scientific approach that examines social behavior and thought in differing contexts can provide that kind of information, and this is one basic reason why social psychologists put their faith in the scientific method: it yields much more conclusive evidence. In fact, as we'll soon see, it is designed to help us determine not just which of the opposite sets of predictions mentioned above is correct, but also when and why one or the other might apply.

But this is not the only reason for being suspicious of common sense. Another one relates to the fact that unlike Mr. Spock of *Star Trek* fame, we are not perfect information-processing machines. On the contrary, as we'll note over and over again (e.g., Chapters 2, 3, 4, and 6), our thinking is subject to several types of biases that can lead us badly astray. Here's one example: Think back over major projects on which you have worked in the past (writing term papers, cooking a complicated dish, painting your room). Now, try to remember two things: (1) your initial estimates about how long it would take you to complete these jobs and (2) how long it actually took. Is there a gap between these two numbers? In all likelihood there is because most of us fall victim to the *planning fallacy*—a strong tendency to believe that projects will take less time than they actually do or, alternatively, that we can accomplish more in a given period of time than is really true. Moreover, we fall victim to this bias in our thought over and over again, despite repeated experiences that tell us "everything takes longer than we think it will." Why are we subject to this kind of error? Research by social psychologists indicates that part of the answer involves a tendency to think about the future when we are estimating how long a job will take. This prevents us from remembering how long similar tasks took in the past and that, in turn, leads us to underestimate the time we will need now (e.g., Buehler, Griffin, & Ross, 1994). This is just one of the many ways in which we can—and often do—make errors in thinking about other people (and ourselves); we'll consider many others in Chapter 3. Because we are prone to such errors in our informal thinking about the social world, we cannot rely on it—or on common sense—to solve the mysteries of social behavior. Rather, we need scientific evidence; and providing such evidence is, in essence, what social psychology is all about.

Social Psychology Focuses on the Behavior of Individuals

Societies differ greatly in terms of their views concerning courtship and marriage, yet it is still individuals who fall in love. Similarly, societies vary greatly in terms of their overall levels of violence, yet it is still individuals who perform aggressive actions or refrain from

doing so. The same argument applies to virtually all other aspects of social behavior, from prejudice to helping: the actions are performed by, and the thoughts occur in, the minds of individuals, although they may, of course, be strongly influenced by other people. Because of this basic fact, the focus in social psychology is strongly on individuals. Social psychologists realize, of course, that we do not exist in isolation from social and cultural influences—far from it. As we will see throughout the book, much social behavior occurs in group settings, and these can exert powerful effects on us. But the field's major interest lies in understanding the factors that shape the actions and thoughts of individuals in social settings.

Social Psychology Seeks to Understand the Causes of Social Behavior and Thought

In a key sense, the heading of this section states the most central aspect of our definition. What it means is that social psychologists are primarily interested in understanding the many factors and conditions that shape the social behavior and thought of individuals—their actions, feelings, beliefs, memories, and inferences concerning other people. Obviously, a huge number of variables play a role in this regard. Most, though, fall under the four major headings described below.

THE ACTIONS AND CHARACTERISTICS OF OTHER PEOPLE Imagine the following events:

You are at a party when you notice that a very attractive person is looking at you and smiling. In fact, this person is looking at you in a way that leaves little room for interpretation: that person is sending a clear signal saying, "Hey, let's get acquainted!"

You are in a hurry and notice that you are driving faster than you usually do—above the speed limit, in fact. Suddenly, up ahead, you see the blinking lights of a state trooper who is in the process of pulling another driver over to the side of the road.

Will these actions by other people have any effect on your behavior and thoughts?

Absolutely. Depending on your own personality, you may blush with pleasure when you see someone looking at you in a "let's get to know each other better" kind of way, and then, perhaps, go over and say "hello." And when you spot the state trooper's blinking light, you will almost certainly slow down—a lot! Instances like these, which occur hundreds of times each day, indicate that other people' behavior often has a powerful impact upon us (see Figure 1.3).



FIGURE 1.3 Reacting to the Actions of Other People
As shown in these scenes, the behavior of other people often exerts powerful effects on our own behavior and thought.

In addition, we are also often affected by others' appearance. Be honest: Don't you behave differently toward highly attractive people than toward less attractive ones? Toward very old people compared to young ones? Toward people who belong to racial and ethnic groups different from your own? And don't you sometimes form impressions of others' personalities and traits from their appearance? Your answer to these questions is probably yes because we do often react to the others' visible

characteristics, such as their appearance (e.g., McCall, 1997; Twenge & Manis, 1998). In fact, research findings (e.g., Hassin & Trope, 2000) indicate that we cannot ignore others' appearance even when we consciously try to do so and, as you probably already guess, it plays an important role in dating and romantic relationships (e.g., Burriss, Roberts, Welling, Puts, & Little, 2011). So despite warnings to avoid "judging books by their covers," we are often strongly affected by other people's appearance—even if we are unaware of such effects and might deny their existence (see Chapter 7). Interestingly, research findings indicate that relying on others' appearance as a guide to their characteristics is not always wrong; in fact, they can be relatively accurate, especially when we can observe others behaving spontaneously, rather than in posed photos (Nauman, Vazire, Rentfrow, & Gosling, 2009).

COGNITIVE PROCESSES Suppose that you have arranged to meet a friend, and this person is late. In fact, after 30 minutes you begin to suspect that your friend will never arrive. Finally, she or he does appear and says, "Sorry…I forgot all about meeting you until a few minutes ago." How will you react? Probably with annoyance. Imagine that instead, however, your friend said, "I'm so sorry to be late. There was a big accident, and the traffic was tied up for miles." Now how will you react? Probably with less annoyance—but not necessarily. If your friend is often late and has used this excuse before, you may be suspicious about whether this explanation is true. In contrast, if this is the first time your friend has been late, or if your friend has never used such an excuse in the past, you may accept it as true. In other words, your reactions in this situation will depend strongly on your memories of your friend's past behavior and your inferences about whether her or his explanation is really true. Situations like this one call attention to the fact that cognitive processes play a crucial role in social behavior and social thought. We are always trying to make sense out of the social world, and this basic fact leads us to engage in lots of social cognition—to think long and hard about other people—what they are like, why they do what they do, how they might react to our behavior, and so on (e.g., Shah, 2003). Social psychologists are well aware of the importance of such processes and, in fact, social cognition is one of the most important areas of research in the field (e.g., Fiske, 2009; Killeya & Johnson, 1998; Swann & Gill, 1997).

ENVIRONMENTAL VARIABLES: IMPACT OF THE PHYSICAL WORLD Are people more prone to wild impulsive behavior during the full moon than at other times (Rotton & Kelley, 1985)? Do we become more irritable and aggressive when the weather is hot and steamy than when it is cool and comfortable (Bell, Greene, Fisher, & Baum, 2001; Rotton & Cohn, 2000)? Does exposure to a pleasant smell in the air make people more helpful to others (Baron, 1997) and does that occur on baseball playing fields as well in crowded and largely unconditioned sections of cities (Larrick, Timmerman, Carton, & Abrevaya, 2011)? Research findings indicate that the physical environment does indeed influence our feelings, thoughts, and behavior, so these variables, too, certainly fall within the realm of modern social psychology.

BIOLOGICAL FACTORS Is social behavior influenced by biological processes and genetic factors? In the past, most social psychologists would have answered no, at least to the genetic part of this question. Now, however, many have come to believe that our preferences, behaviors, emotions, and even attitudes are affected, to some extent, by our biological inheritance (Buss, 2008; Nisbett, 1990; Schmitt, 2004), although social experiences too have a powerful effect, and often interact with genetic factors in generating the complex patterns of our social lives (e.g., Gillath, Shaver, Baek, & Chun, 2008).

The view that biological factors play an important role in social behavior comes from the field of **evolutionary psychology** (e.g., Buss, 2004; Buss & Shackelford, 1997). This new branch of psychology suggests that our species, like all others on the planet, has been subject to the process of biological evolution throughout its history, and that as a result of this process, we now possess a large number of evolved psychological mechanisms that help (or once helped) us to deal with important problems relating to survival. How do these become

evolutionary psychology
A new branch of psychology that seeks to investigate the potential role of genetic factors in various aspects of human behavior.

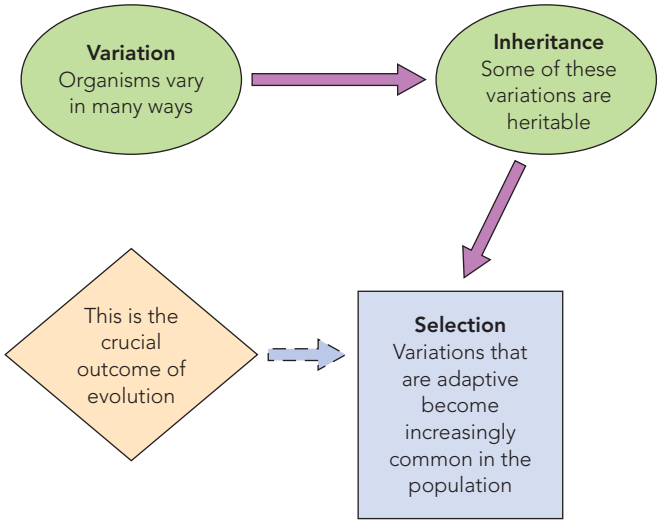


FIGURE 1.4 Evolution: An Overview
As shown here, evolution involves three major components: variation, inheritance, and selection.

part of our biological inheritance? Through the process of evolution, which, in turn, involves three basic components: *variation*, *inheritance*, and *selection*. Variation refers to the fact that organisms belonging to a given species vary in many different ways; indeed, such variation is a basic part of life on our planet. Human beings, as you already know, come in a wide variety of shapes and sizes, and vary on what sometimes seems to be an almost countless number of dimensions.

Inheritance refers to the fact that some of these variations can be passed from one generation to the next through complex mechanisms that we are only now beginning to fully understand. Selection refers to the fact that some variations give the individuals who possess them an "edge" in terms of reproduction: they are more likely to survive, find mates, and pass these variations on to succeeding generations. The result is that over time, more and more members of the species possess these variations. This change in the characteristics of a species over time—immensely long periods of time—is the concrete outcome of evolution. (See Figure 1.4 for a summary of this process.)

Social psychologists who adopt the evolutionary perspective suggest that this process applies to at least some aspects of social behavior. For instance, consider the question of mate preference. Why do we find some people attractive? According to the evolutionary perspective, because the characteristics they show—symmetrical facial features; well-toned, shapely bodies; clear skin; lustrous hair—are associated with "good genes"—they suggest that the people who possess them are likely to be healthy and vigorous, and therefore good mates (e.g., Schmitt & Buss, 2001; Tesser & Martin, 1996). For instance, these characteristics—the ones we find attractive—indicate that the people who show them have strong immune systems that protect them from many illnesses (e.g. Burriss et al., 2011; Li & Kenrick, 2006). Presumably, a preference for characteristics associated with good health and vigor among our ancestors increased the chances that they would reproduce successfully; this, in turn, contributed to our preference for people who possess these aspects of appearance.

Here's another example, and one that is perhaps a bit more surprising. When asked to indicate the characteristics in potential romantic partners that they find desirable, both genders—but especially women—rate a sense of humor high on the list (e.g., Buss, 2008). Why? From an evolutionary point of view, what is it about humor that makes it a desirable characteristic in others? One possibility is that a sense of humor signals high intelligence, and this tends to make humorous people attractive—after all, they have good genes (e.g., Griskevicius et al., in press). But another possibility is that a sense of humor signals something else: interest in forming new relationships. In other words, it is a sign that the humorous person is available—and interested. Research by Li et al. (2009) found that people are more likely to use humor and laugh at humor by others when they find these people attractive than when they do not, and that they perceived people who used humor during speed dating sessions as showing more romantic interest than ones who did not (see Figure 1.5).

Other topics have been studied from the evolutionary perspective (e.g., helping others; aggression; preferences for various ways of attracting people who are already in a relationship), and we'll describe this research in other chapters. Here, however, we wish to emphasize the fact that the evolutionary perspective does not suggest that we inherit specific patterns of social behavior; rather, it contends that we inherit tendencies or predispositions that may be apparent in our overt actions, depending on the environments in which we live. Similarly, this perspective does not suggest that we are "forced" or driven by our genes to act in specific ways. Rather, it merely suggests that because of our genetic inheritance, we have tendencies to behave in certain ways that, at least in the past, enhanced the chances that our ancestors would survive and pass their genes

on to us. These tendencies can be—and often are—overridden by cognitive factors and the effects of experience (i.e., learning; Pettijohn & Jungeberg, 2004). For instance, what is viewed as attractive changes over time and is often very different in diverse cultures (e.g., overweight women are particularly desirable in Nigeria but less so in contemporary North America). So yes, genetic factors play some role in our behavior and thought, but they are clearly only one factor among many that influence how we think and act.



FIGURE 1.5 Humor: An Important "Plus" in Dating
Research findings indicate that humor is viewed as a desirable charactersitic in potential romantic partners, partly because it is perceived as a sign that the person demonstrating it is interested in forming a new relationship. Such effects occur in many situations, including speed dating, as shown here. So, if you want romantic partners, keep on smiling and make jokes!

The Search for Basic Principles in a Changing Social World

One key goal of science is the development of basic principles that are accurate regardless of when or where they are applied or tested. For instance, in physics, Einstein's equation $e = mc^2$ is assumed to be true everywhere in the universe, and at all times—now, in the past, and in the future. Social psychologists, too, seek such basic principles. While they don't usually develop elegant mathematical expressions or equations, they do want to uncover the basic principles that govern social life. For instance, they'd like to determine what factors influence attraction, helping, prejudice, first impressions of other people, and so on. And the research they conduct is designed to yield such knowledge—basic principles that will be true across time and in different cultures.

On the other hand, they recognize the fact that cultures differ greatly and that the social world in which we live is constantly changing—in very important ways. For instance, even today, cultures vary greatly with respect to when and where people are expected to "dress up" rather than dress casually. While casual is acceptable in almost all contexts in the United States, more formal "dressy" attire is still expected in other cultures. This is a relatively trivial example, but the same point applies to more important aspects of social life, too: Should teenagers be allowed to date and meet without adult supervision? At what age should marriage occur? Are "gifts" to public officials acceptable or illegal bribes (see Figure 1.6)? At what age should people retire, and how should they be treated after they do? Cultures differ tremendously in these and countless other ways, and this complicates the task of establishing general principles of social behavior and social thought.



FIGURE 1.6 Cultures Differ in Many Ways—Including Their Views About Bribes
In some cultures, it is considered acceptable—or even essential—to offer gifts (bribes?) to public officials. In others, such actions will land you in jail!

In addition, the social world is changing—and very rapidly, too. Because of social networks, cell phones, online dating, and many other changes, people now meet potential romantic partners in different ways than in the

past when, typically, they were introduced by friends or met at dances arranged by their schools, churches, or other social organizations. Does this mean that the foundations of attraction are different today than in the past? Social psychologists believe that despite these changes, the same basic principles apply: Physical attractiveness is still a basic ingredient in romance, and although influence is now exerted in many ways not possible in the past (e.g., pop-ads on the Internet), the basic principles of persuasion, too, remain much the same (Goel, Mason, & Watts, 2010). In short, although the task of identifying basic, accurate principles of social behavior and social thought is complicated by the existence of huge cultural differences and rapid changes in social life, the goals of social psychological research remain within reach: uncovering basic, accurate facts about the social side of life that do apply in a wide range of contexts and situations.



KEY POINTS

- Social psychology is the scientific field that seeks to understand the nature and causes of individual behavior and thought in social situations.
- It is scientific in nature because it adopts the values and methods used in other fields of science.
- Social psychologists adopt the scientific method because "common sense" provides an unreliable guide to social behavior, and because our personal thought is influenced by many potential sources of bias.
- Social psychology focuses on the behavior of individuals, and seeks to understand the causes of social behavior and thought, which can involve the behavior and appearance of others, social cognition, environmental factors, cultural values, and even biological and genetic factors.
- Social psychology seeks to establish basic principles of social life that are accurate across huge cultural differences and despite rapid and major changes in social life.
- Important causes of social behavior and thought include the behavior and characteristics of other people, cognitive processes, emotions, cultures, and genetic factors.

Social Psychology: Summing Up

In sum, social psychology focuses mainly on understanding the causes of social behavior and social thought—on identifying factors that shape our feelings, behavior, and thought in social situations. It seeks to accomplish this goal through the use of scientific methods, and it takes careful note of the fact that social behavior and thought are influenced by a wide range of social, cognitive, environmental, cultural, and biological factors.

The remainder of this text is devoted to describing some of the key findings of social psychology. This information is truly fascinating, so we're certain that you will find it of interest—after all, it is about *us* and the social side of our lives! We're equally sure, however, that you will also find the outcomes of some research surprising, and that it will challenge many of your ideas about people and social relations. So please get ready for some new insights. We predict that after reading this book, you'll never think about the social side of life in quite the same way as before.

Social Psychology: Advances at the Boundaries

Textbooks, unlike fine wine, don't necessarily improve with age. So, to remain current, they must keep pace with changes in the fields they represent. Making certain that this book is current, in the best sense of this term, is one of our key goals, so you can be sure that what's presented in the chapters that follow provides a very contemporary summary of our current

knowledge of the social side of life. Consistent with this belief, we now describe several major trends in modern social psychology—themes and ideas that represent what's newest and at the center of our field's attention. We do this primarily to emphasize the broad scope of social psychology, and also to alert you to topics we consider again in later chapters.

Cognition and Behavior: Two Sides of the Same Social Coin

In the past (actually, what's getting to be the dim and distant past!), social psychologists could be divided into two distinct groups: those who were primarily interested in social *behavior*—how people act in social situations—and those who were primarily interested in social *cognition*—how people attempt to make sense out of the social world and to understand themselves and others. This division has now totally disappeared. In modern social psychology, behavior and cognition are seen as intimately, and continuously, linked. In other words, there is virtually universal agreement in the field that we cannot hope to understand how and why people behave in certain ways in social situations without considering their thoughts, memory, intentions, emotions, attitudes, and beliefs. Similarly, virtually all social psychologists agree that there is a continuing and complex interplay between social thought and social behavior. What we think about others influences our actions toward them, and the consequences of these actions then affect our social thought. So, the loop is continuous and in trying to understand the social side of life, modern social psychology integrates both. That is be our approach throughout the book, and it is present in virtually every chapter.

The Role of Emotion in the Social Side of Life

Can you imagine life without feelings—emotions or moods? Probably not, because this, too, is a very central aspect of social life—and life more generally. Social psychologists have always been interested in emotions and moods, and with good reason: they play a key role in many aspects of social life. For instance, imagine that you want a favor from a friend or acquaintance—when would you ask for it, when this person is in a good mood or a bad one? Research findings indicate that you would do much better when that person is in a good mood, because positive moods (or *affect*, as social psychologists term such feelings) do increase our tendency to offer help to others (e.g., Isen & Levin, 1972). Similarly, suppose you are meeting someone for the first time. Do you think your current mood might influence your reactions to this person? If you answered "yes," you are in agreement with the results of systematic research, which indicates our impressions of others (and our thoughts about them) are strongly influenced by our current moods. More recently, social psychologists have been investigating the role of moods in a wider range of social behaviors and social thought (e.g., Forgas, Baumeister, & Tice, 2009). Overall, interest in this topic, including the impact of specific emotions, has increased. So, we include it here as another area in which rapid advances are being made at the boundaries of our current knowledge of social life. In addition, we represent this interest throughout the book in special sections within each chapter (e.g., "Emotion and Attitudes," "Emotion and Helping," "Emotion and Social Cognition"), so be on the lookout for these sections because they report some of the most fascinating research currently occurring in our field.

Relationships: How They Develop, Change, and Strengthen—or End

If the social side of life is as important as we suggested at the start of this chapter—and we firmly believe that it is—then **relationships** with others are its building blocks. When they are successful and satisfying, they add tremendously to our happiness, but when they go "wrong," they can disrupt every other aspect of our lives, and undermine our psychological health and well-being, and even our own self-concept (e.g., Slotter, Gardner, & Finkel,

relationships
Our social ties with other persons, ranging from casual acquaintance or passing friendships, to intense, long-term relationships such as marriage or lifetime friendships.

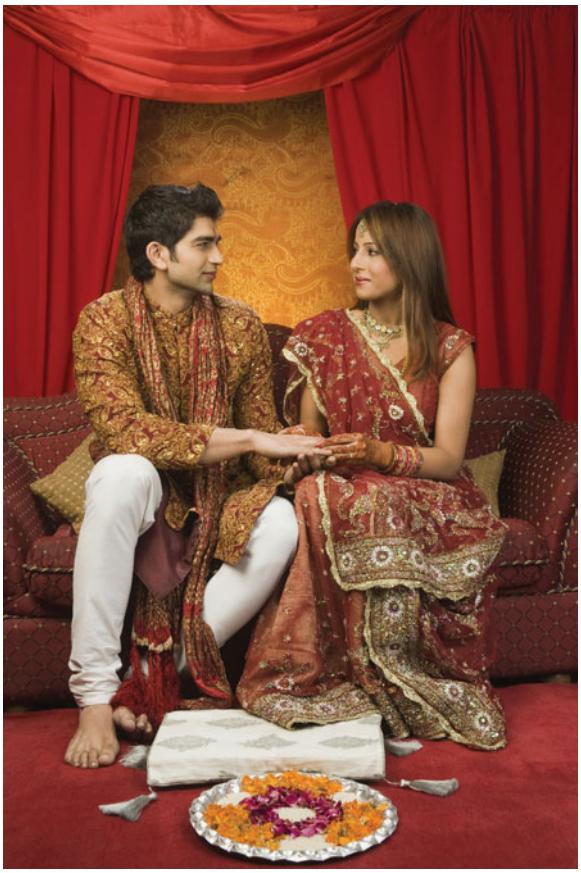


FIGURE 1.7 The Warm Glow of Love
When couples are in love, they often perceive each other in unreaslitically favorable ways. Is that good or bad for their future relationships? The answer is complex, but reaserch findings indicate that as long as they show some degree of reality or accuracy, it may be beneficial.

2010). Given these basic facts, social psychologists have long sought to understand the nature of social relationships—how they begin and change over time, and why, gradually, some strengthen and deepen, while others weaken and die—often, after causing tremendous pain to the people involved. In recent years, however, interest in these topics has increased greatly, and relationships are now receiving more research attention than ever before. The results of this research have been—and continue to be—remarkably revealing. We consider relationships in detail in Chapter 7, but here, to give you the flavor of this growing body of knowledge, we mention just a couple of lines of important and revealing research.

One such topic relates to the following question: "Is it better, in terms of building a strong relationship, to view one's partner (boyfriend, girlfriend, or spouse) realistically, or as we often do, through a 'golden, positive glow'?" Folklore suggests that "love is blind," and when in love, many people do tend to see only good in their partners (see Figure 1.7). Is that tendency good or bad for their relationships? Research findings suggest that in general, it is good, but only if it is restrained by a healthy degree of reality (i.e., accuracy; e.g., Fletcher, Simpson, & Boyes, 2006). For example, in one study on this issue (e.g., Luo & Snider, 2009), several hundred newlywed couples were asked to complete measures that revealed the extent to which they perceived their new spouses accurately, in a positive light, and as similar to themselves in many ways. Accuracy was measured by comparing each spouse's ratings of their partner on many dimensions with their partner's own self-ratings. The closer these scores, the higher the accuracy. Similarity bias was measured in a parallel way in terms of the extent to which each partner perceived his or her spouse as more similar to themselves than was actually the case. These measures of accuracy, positivity bias, and similarity bias were then related to marital satisfaction as expressed by both partners in each couple. Results revealed a clear picture: all three dimensions were important in predicting marital satisfaction. Positive and similarity bias contributed to such happiness, but accuracy did too. Overall, these findings indicate that it is indeed good to hold favorable perceptions of our romantic partners, but that these must be moderated by a dash of accuracy, too. We return to these questions in Chapter 7; here, we merely mention them to give you a basic idea of the kind of questions investigated in the context of relationships.

Another question concerning relationships that has received growing attention from social psychologists is this: What are the effects of a breakup? This is a case where common sense offers contradictory answers. On the one hand, it is widely believed that the breakup of a romantic relationship is traumatic, and may leave lasting psychological scars behind. On the other, the saying "What doesn't kill you makes you stronger" suggests that there are actual benefits from such painful experiences. Research on breakups suggests that there is some truth in both views. On the one hand, the breakup of romantic relationships *is* painful and distressing; in fact, it has been found to negatively affect individuals' self-concept, so that, for instance, they feel more vulnerable and less certain about who, precisely, they are (i.e., the clarity of their self-concept is reduced; Slotter et al., 2010). On the other hand, it appears that experiencing a breakup may increase the desire for another relationship, and encourage the people involved to actually form new ones—"on the rebound" (Spielmann, MacDonald, & Wilson, 2009). While there are real risks involved in rapidly forming new relationships, they do offer at least one major benefit: they help the people involved to let go of their former relationship and "move on" with their lives. These benefits are especially strong for people who are high in

what social psychologists term *anxious attachment*—anxiety over the possibility of losing a partner and/or an inability to get as close to partners as one would prefer.

Overall, research on relationships has provided many important insights into this crucial part of our social lives, and offers helpful suggestions on how they can be strengthened and developed so that their beneficial effects are maximized and their potential costs reduced.

Social Neuroscience: Where Social Psychology and Brain Research Meet

In a basic sense, everything we do, feel, imagine, or create reflects activity within our brains. Are you understanding the words on this page? If so, it is the result of activity in your brain. Are you in a good mood? A bad one? Whatever you are feeling also reflects activity in your brain and biological systems. Can you remember what your third-grade teacher looked like? What your first ride on a roller coaster felt like? The smell of your favorite food? Do you have plans for the future—and do you think they can actually be achieved? All of these events and processes are the result of activity in various areas of your brain. In the past 20 years, powerful new tools for measuring activity in our brains as they function have been developed: functional magnetic resonance imaging (fMRI), positron emission tomography (PET) scans, and other techniques. Although they were initially developed for medical uses, and have generated major advances in surgery and other branches of medicine, they have also allowed psychologists and other scientists to peer into the human brain as people engage in various activities, and so to find out just what's happening at any given time. The result is that we now know much more about the complex relationships between neural events and psychological ones—feelings, thoughts, and overt actions.

Social psychologists, too, have begun to use these new tools to uncover the foundations of social thought and social behavior in our brains—to find out what portions of the brain and what complex systems within it are involved in key aspects of our social life—everything from prejudice and aggression, through underperforming on tasks due to "choking under pressure" (Mobbs et al., 2009), and empathy and helping (e.g., Van Berkum, Hollmean, Nieuwaland, Otten, & Murre, 2009). In conducting such research, social psychologists use the same basic tools as other scientists—they study events in the brain (through the use of fMRI and other kinds of brain scans), other neural activity, and even changes in the immune system (e.g., Taylor, Lerner, Sherman, Sage, & McDowell, 2003) in order to determine how these events are related to important social processes. The findings of this research have been truly fascinating. Here's one example of what we mean.

Attitudes and values are an important part of the social side of life; as we'll see in Chapter 5, they often shape our overt behavior and underlie powerful emotional reactions to events and people. But how are they represented in the brain, and how do they exert their powerful effects on our behavior, thought, and emotions? Social neuroscience research is providing intriguing answers. For example, consider a study by Van Berkum and colleagues (2009). This investigation was designed to determine what happens in the brain when people encounter statements that are consistent or inconsistent with their strongly held values and attitudes. To do this, they recruited two groups of participants known to hold opposite views on many social issues. One group (members of a strict Christian church) were known to be against euthanasia, growing equality of women in society, abortion, and the use of drugs. The other, self-described as "nonreligious," held opposite views on all these issues. Both groups were then exposed to statements relating to these attitudes on a computer screen, and while viewing them, electrical activity in their brains was carefully recorded. A key question asked by the researchers was, How quickly do people react, in terms of brain activity, to statements that disagree with their own attitudes or values? Do they react this way as soon as they encounter a single word inconsistent with their views (e.g., "acceptable" in the statement "I think euthanasia is acceptable…" if they are against this action) or only after reading the entire statement and considering it carefully. Previous research indicated that certain patterns of

activity (N400, one kind of *event-related potential*—a kind of activity in the brain), occur very quickly when individuals encounter words inconsistent with their values—only 250 milliseconds after seeing them—and indicate that intensified processing of this word is occurring. Other patterns, in contrast, occur somewhat later, and reflect negative reactions to the value-inconsistent statement. It was predicted that each group would show stronger N400 reactions to words that were inconsistent with their values, so that, for instance, the Christian group would show stronger reactions to the word "acceptable" in connection with euthanasia, while the other group would express stronger reactions to the word "unacceptable" when linked to euthanasia. Results offered strong support for these predictions, and suggest that we do indeed process information that disagrees with our attitudes or values very quickly—long before we can put such reactions into words. So yes, attitudes and values do indeed exert powerful and far-reaching effects on activity within our brains—and on our overt actions.

Here's another example of how social psychologists are using the tools of neuroscience to study important aspects of social thought and behavior. Have you ever heard of *mirror neurons*? They are neurons in our brains that are activated during the observation and execution of actions, and it has been suggested that they play a key role in *empathy*—our capacity to experience, vicariously, the emotions and feelings of other people (e.g., Gazzola, Aziz-Zadeh, & Keysers, 2006). Mirror neurons are located in a portion of the brain known as the *frontal operculum* and in an intriguing study, Montgomery, Seeherman, and Haxby (2009) suggested that perhaps people who score high on a questionnaire measuring empathy would show more activity in this area of their brains when they viewed social facial expressions shown by others. To test this prediction, the researchers exposed two groups of individuals—ones who had scored high in a measure of empathy or low on this measure (an index of the capacity to take the perspective of other people) to video clips of others' facial expressions (e.g., smiling, frowning) or to faces that showed nonsocial movements (i.e., movements not associated with particular emotions). Activity in the brains of both groups of participants was recorded through fMRI scans as they watched the videos. Results were clear: as predicted, people high or moderate in empathy did indeed show higher activity in the frontal operculum (where mirror neurons are located) than people low in empathy (see Figure 1.8).

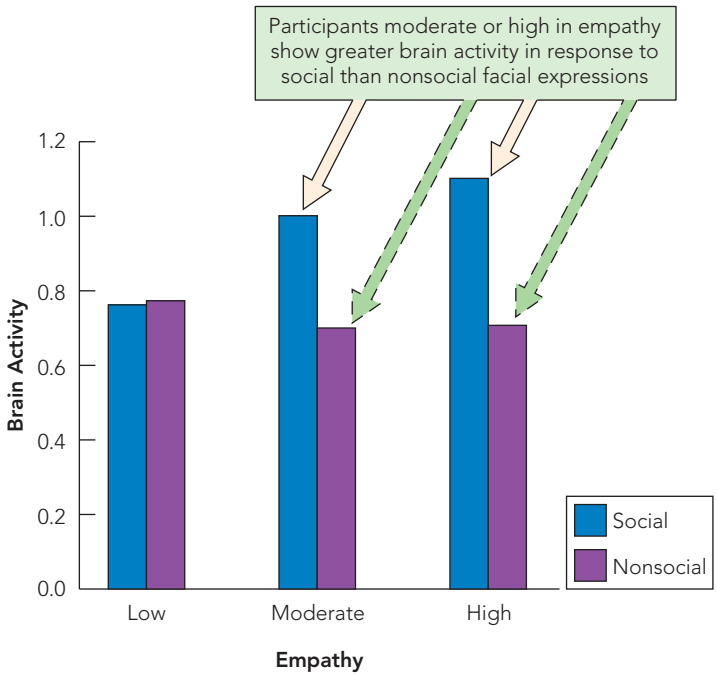


FIGURE 1.8 The Neural Basis of Empathy

Individuals high or moderate in a measure of empathy (the capacity to see the world through others' eyes) showed more activity in a portion of their brains (the frontal operculum) than persons low in empathy, when watching videos of other persons showing social facial expressions. In contrast, the groups did not differ in brain activity while watching videos showing nonsocial facial movements (i.e., ones unrelated to emotions). (Source: Based on data from Montgomery, Seeherman, & Haxby, 2009).

Research in the rapidly expanding field of social neuroscience is clearly at the forefront of advances in social psychology, and we represent it fully—and often—in this text. We should insert one warning, however. As noted by several experts in this field (e.g., Cacioppo et al., 2003), social neuroscience cannot provide the answer to every question we have about social thought or behavior. There are many aspects of social thought that cannot easily be related to activity in specific areas of the brain—aspects such as attitudes, attributions, group identities, and reciprocity (e.g., Willingham & Dunn, 2003). In principle, all of these components of social thought reflect activity in the brain, but this does not necessarily mean that it is best to try to study them in this way. In fact, the situation may be similar to that existing between chemistry and physics. All chemists agree that ultimately, every chemical reaction can be explained in terms of physics.

But the principles of chemistry are still so useful that chemists continue to use them in their research and do not all rush out and become physicists. The same may well be true for social psychology: it does not have to seek to understand all of its major topics in terms of activities in the brain or nervous system; other approaches, which we describe in later chapters, are still useful and can provide important new insights. Throughout this book, therefore, we describe research that uses a wide range of methods, from brain scans on the one hand, to direct observations of social behavior on the other. This reflects the current, eclectic nature of social psychology and is, therefore, the most appropriate content for this book.

The Role of Implicit (Nonconscious) Processes

Have you ever had the experience of meeting someone for the first time and taking an immediate liking—or disliking—to that person? Afterward, you may have wondered, "Why do I like (dislike) this person?" But probably, you didn't wonder for long because we are all experts at finding good reasons to explain our own actions or feelings. This speed in no way implies that we really do understand why we behave or think in certain ways. And in fact, a growing theme of recent research in social psychology has been just this: in many cases we really don't know why we think or behave as we do in social contexts. And, partly because of our errors in the way we process social information, and partly because we change greatly over time, we don't even know—with clarity—what would make us happy (Gilbert, 2006). So, for instance, people get a tattoo that they think will make them happy, only to realize, years later, that it is making them unhappy, not happy. In addition, our thoughts and actions are shaped by factors and processes of which we are only dimly aware, at best, and which often take place in an automatic manner, without any conscious thought or intentions on our part. This is one more reason why social psychologists are reluctant to trust "common sense" as a basis for reliable information about social behavior or social thought: We are unaware of many of the factors that influence how we think and how we behave and so cannot report on them accurately (e.g., Pelham, Mirenberg, & Jones, 2002). For example, consider first impressions: Recent findings indicate that we form these incredibly quickly—often within mere seconds of meeting other people (e.g., Gray, 2008). And, amazingly, sometimes these impressions appear to be accurate: We can form valid impressions of others' personalities even from a very brief exposure to them (e.g., Carney, Colvin, & Hall, 2007). But the picture is a mixed one: sometimes these first impressions are accurate and sometimes they are very wrong. This raises another question: Can we tell when our first impressions are likely to be useful and when they are not? In other words, can we tell whether to have confidence in them or mistrust them? Recent evidence reported by Ames, Kammrath, Suppes, and Bolger (2010) indicates that we cannot: We can't intuit when these impressions are likely to be accurate and when they are not. So, as these authors suggest (p. 273), "snap impression accuracy is sometimes above chance…" but we can't tell when that is the case. Clearly, nonconscious processes influence our judgments and actions in such cases, but perhaps they should not.

Research on the role of implicit (nonconscious) processes in our social behavior and thought has examined many other topics, such as the impact of our moods on what we tend to remember about other people or complex issues (e.g., Ruder & Bless, 2003), how negative attitudes toward members of social groups other than our own that we deny having can still influence our reactions toward them (e.g., Fazio & Hilden, 2001), and how we automatically evaluate people belonging to various social groups once we have concluded that they belong to that group (Castelli, Zobmaister, & Smith, 2004). In short, nonconscious factors and processing seem to play an important role in many aspects of social thought and social behavior. We examine such effects in several chapters since they continue to represent an important focus of current research (see, e.g., Chapters 2 and 6).



FIGURE 1.9 Diversity: A Fact of Life in Many Countries in the 21st Century
Populations in many countries—including the United States—are becoming increasingly ethnically diverse. Social psychologists take careful account of this fact by conducting research focused on understanding the role of cultural factors in social behavior and social thought.

Taking Full Account of Social Diversity

There can be no doubt that the United States—like many other countries—is undergoing a major social and cultural transformation. Recent figures indicate that 64 percent of the population identifies itself as White (of European heritage), while fully 36 percent identifies itself as belonging to some other group (13 percent African American, 4.5 percent American Indian, 14 percent Hispanic, 4.5 percent Asian/Pacific Islander, and 7 percent some other group). This represents a tremendous change from the 1960s, when approximately 90 percent of the population was of European descent. Indeed, in several states (e.g., California, New Mexico, Texas, Arizona), people of European heritage are now a minority (see Figure 1.9). In response to these tremendous shifts, psychologists have increasingly recognized the importance of taking cultural factors and differences into careful account in everything they do—teaching, research, counseling, and therapy; and social psychologists are certainly no exception to this rule. They have been increasingly sensitive to the fact that individuals' cultural, ethnic, and racial heritage often play a key role in their self-identity, and that this, in turn, can exert important effects on their behavior. This is in sharp contrast to the point of view that prevailed in the past, which suggested that cultural, ethnic, and gender differences are relatively unimportant. In contrast to that earlier perspective, social psychologists currently believe that such differences are very important, and must be taken carefully into account in our efforts to understand human behavior. As a result, psychology in general, and social psychology as well, now adopts a **multicultural perspective—**one that carefully and clearly recognizes the potential importance of gender, age, ethnicity, sexual orientation, disability, socioeconomic status, religious orientation, and many other social and cultural dimensions. This perspective has led to important changes in focus of social psychological research, which we cover in-depth in Chapters 4 and 6, and this trend seems likely to continue.

For instance, consider a study conducted in 10 different countries around the world, focused on what kind of body shape both men and women find most attractive in women (Swami et al., 2010). Participants were shown the drawings in Figure 1.10, and asked to choose the one they found most attractive; women were asked to select the one that they thought would be most attractive to men of their own age, and the one that most closely matched their current body. Results indicated that there were indeed cultural differences in the ratings provided by participants: raters in Oceania, south and west Asia, and Southeast Asia preferred heavier body types then those in North America and east Asia. However, larger differences occurred within cultures in terms of socioeconomic status: higher SES people (i.e., those higher in education and income) preferred slimmer body builds to those of lower SES status. This suggests that large differences exist with respect to this very basic aspect of social perception within cultures as well as between them. Clearly, increased recognition of diversity and cuiltural differences is a hallmark of

multicultural perspective
A focus on understanding the cultural and ethnic factors that influence social behavior.

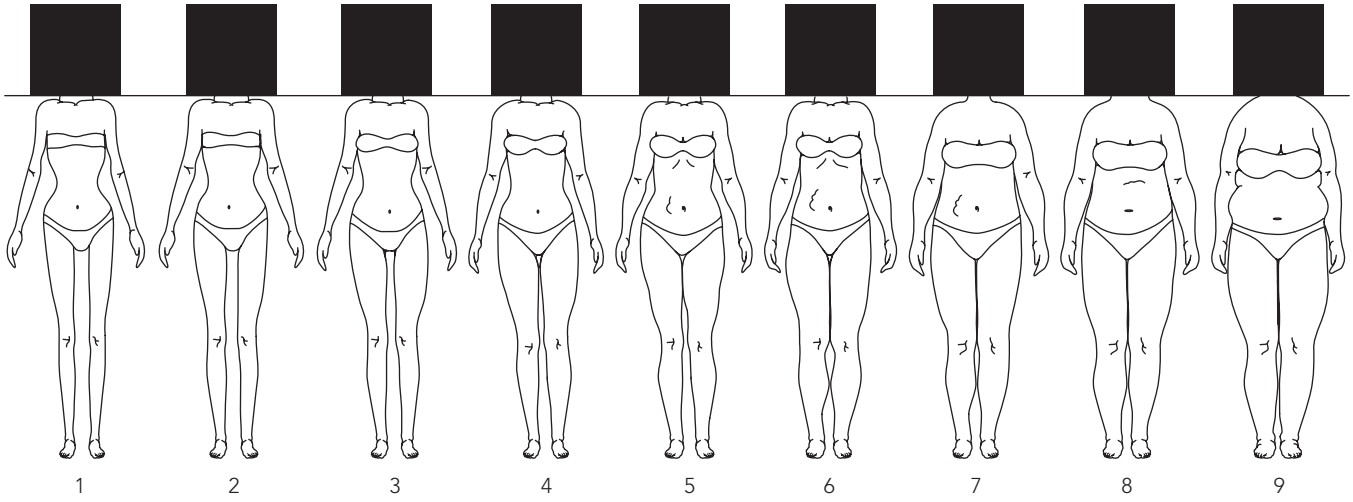


FIGURE 1.10 Cultural Differences in Preferred Body Types

Do people in different cultures prefer different body types or weights in women? Research conducted in 10 different countries indicates that they do, with people from cultures in some parts of Asia and Europe preferring rounder figures than people in North America. However, within each culture, differences between people high and low in socioeconomic status are even greater than those between different cultures. *(Source: V. Swami, et.al, PERSONALITY AND SOCIAL PSYCHOLOGY BULLETIN, 36 (3) March 2010, p.17. © 2010 Sage Publications. Reprinted by permissions of SAGE Publications.).*

modern social psychology, and we discuss research highlighting the importance of such factors at many points in this book.

KEY POINTS

- Social psychologists currently recognize that social thought and social behavior are two sides of the same coin, and that there is a continuous, complex interplay between them.
- There is growing interest among social psychologists in the role of emotion in social thought and social behavior.
- The formation and development of **relationships** is another major trend in the field.
- Yet another major trend involves growing interest in social neuroscience—efforts to relate activity in the brain to key aspects of social thought and behavior.
- Our behavior and thought is often shaped by factors of which we are unaware. Growing attention to such implicit (nonconscious) processes is another major theme of modern social psychology.
- Social psychology currently adopts a **multicultural perspective.** This perspective recognizes the importance of cultural factors in social behavior and social thought, and notes that research findings obtained in one culture do not necessarily generalize to other cultures.

How Social Psychologists Answer the Questions They Ask: Research as the Route to Increased Knowledge

Now that we've provided you with an overview of some of the current trends in social psychology, we can turn to the third major task mentioned at the start of this chapter: explaining how social psychologists attempt to answer questions about social behavior and social thought. Since social psychology is scientific in orientation, they usually seek to accomplish this task through systematic research. To provide you with basic information

about the specific techniques they use, we examine three related topics. First, we describe basic methods of research in social psychology. Next, we consider the role of theory in such research. Finally, we touch on some of the complex ethical issues relating to social psychological research.

Systematic Observation: Describing the World Around Us

One basic technique for studying social behavior involves **systematic observation**—carefully observing behavior as it occurs. Such observation is not the kind of informal observation we all practice from childhood on, such as people watching in an airport; rather, in a scientific field such as social psychology it is observation accompanied by careful, accurate measurement of a particular behavior across people. For example, suppose that a social psychologist wanted to find out how frequently people touch each other in different settings. The researcher could study this topic by going to shopping malls, restaurants and bars, college campuses, and many other locations and observe, in those settings, who touches whom, how they touch, and with what frequency. Such research (which has actually been conducted; see Chapter 3), would be employing what is known as *naturalistic observation*—observation of people's behavior in natural settings (Linden, 1992). Note that in such observation, the researcher would simply record what is happening in each context; she or he would make no attempt to change the behavior of the people being observed. In fact, such observation requires that the researcher take great pains to avoid influencing the people observed in any way. Thus, the psychologist would try to remain as inconspicuous as possible, and might even try to hide behind natural barriers such as telephone poles, walls, or even bushes!

Another technique that is often included under the heading of systematic observation is known as the **survey method**. Here, researchers ask large numbers of people to respond to questions about their attitudes or behavior. Surveys are used for many purposes—to measure attitudes toward specific issues such as smoking, to find out how voters feel about various political candidates, to determine how people feel about members of different social groups, and even to assess student reactions to professors (your college or university probably uses a form on which you rate your professors each semester). Social psychologists often use this method to assess attitudes toward a variety of social issues—for instance, national health care reform or affirmative action programs. Scientists and practitioners in other fields use the survey method to measure everything from life satisfaction around the globe to consumer reactions to new products.

Surveys offer several advantages. Information can be gathered about thousands or even hundreds of thousands of people with relative ease. In fact, surveys are now often conducted online, through the Internet. For instance, recent research on personal happiness is being conducted this way. To see for yourself how it works, just visit www.authentichappiness.com. The surveys presented there have been prepared by famous psychologists, and your replies—which are entirely confidential—will become part of a huge data set that is being used to find out why people are happy or unhappy, and ways in which they can increase their personal satisfaction with life. The site has been visited by millions of people and currently has over 750,000 registered users! (We'll return to this topic in detail in Chapter 12). In addition, survey sites can be used for many other purposes—for instance, to see how students rate their professors (see Figure 1.11).

systematic observation
A method of research in which behavior is systematically observed and recorded.

survey method
A method of research in which a large number of people answer questions about their attitudes or behavior.

In order to be useful as a research tool, though, surveys must meet certain requirements. First, the people who participate must be representative of the larger population about which conclusions are to be drawn—which raises the issue of sampling. If this condition is not met, serious errors can result. For instance, suppose that the website shown in Figure 1.11 is visited only by people who are already very happy—perhaps because unhappy people don't want to report on their feelings. Any results obtained would be questionable for describing American levels of happiness, because they do not represent

the entire range of happiness in the population as a whole.

Yet another issue that must be carefully addressed with respect to surveys is this: The way in which the items are worded can exert strong effects on the outcomes obtained. For instance, continuing with the happiness example we have been using, suppose a survey asked people to rate, "How happy are you in your life right now?" (on a 7-point scale where 1 = very unhappy and 7 = very happy). Many people (most?) might well answer 4 or above because overall, most people do seem to be relatively happy much of the time. But suppose the question asked: "Compared to the happiest you have ever been, how happy are you right now in your life?" (1 = much less happy; 7 = just as happy). In the context of this comparison to your peak level of happiness, many people might provide numbers lower than 4, because they know they have been happier *sometime* in the past. Comparing the results from these questions could be misleading, if the differences between them were ignored.

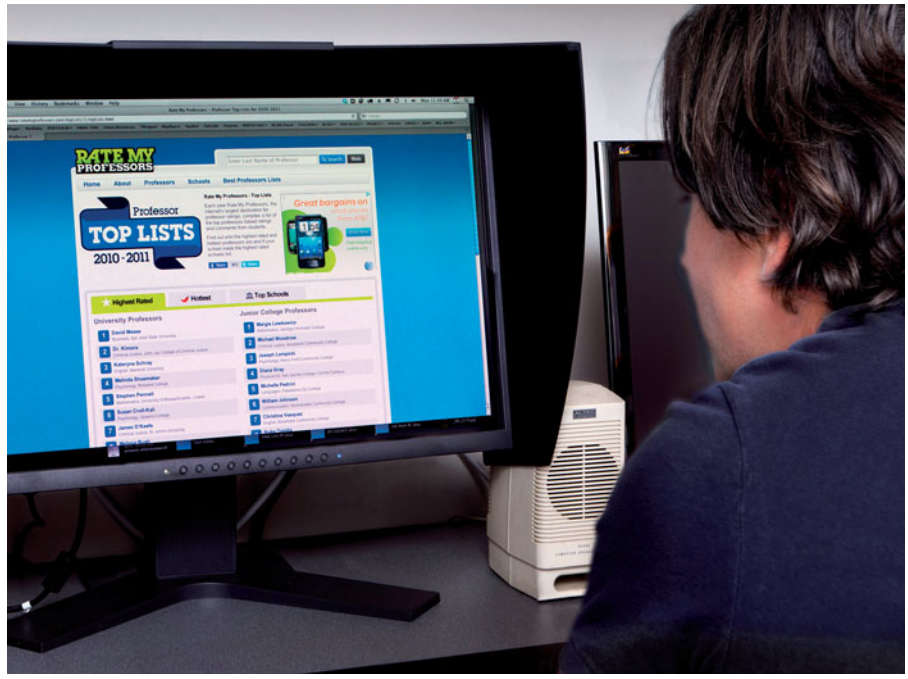


FIGURE 1.11 Using the Internet to Conduct Research—Or Just to Find Out How Other Students Rate Your Professor

Social psychologists sometimes collect survey data from sites they establish on the Internet. Many of these are set up for a specific study, but others, like the one shown here, remain open permanently, and often provide data from hundreds of thousands of persons. In addition, survey sites can be used for many other purposes—for instance, to learn how other students rate your professors.

In sum, the survey method can be a useful approach for studying some aspects of social behavior, but the results obtained are accurate only to the extent that issues relating to sampling and wording are carefully addressed.

Correlation: The Search for Relationships

At various times, you have probably noticed that some events appear to be related to the occurrence of others: as one changes, the other changes, too. For example, perhaps you've noticed that people who drive new, expensive cars tend to be older than people who drive old, inexpensive ones, or that people using social networks such as Facebook tend to be relatively young (although this is changing somewhat now). When two events are related in this way, they are said to be correlated, or that a correlation exists between them. The term *correlation* refers to a tendency for one event to be associated with changes in the other. Social psychologists refer to such changeable aspects of the natural world as *variables*, since they can take different values.

From a scientific point of view, knowing that there is a correlation between two variables can be very useful. When a correlation exists, it is possible to predict one variable from information about one or more other variables. The ability to make such predictions is one important goal of all branches of science, including social psychology. Being able to make accurate predictions can be very helpful. For instance, imagine that a correlation is observed between certain attitudes on the part of individuals (one variable) and the likelihood that they will later be very difficult to work with, both for their coworkers and boss (another variable). This correlation could be very useful in identifying potentially

dangerous people so that companies can avoid hiring them. Similarly, suppose that a correlation is observed between certain patterns of behavior in married couples (e.g., the tendency to criticize each other harshly) and the likelihood that they will later divorce. Again, this information might be helpful in counseling the people involved and perhaps, if this was what they desired, in saving their relationship (see Chapter 7 for a discussion of why long-term relationships sometimes fail).

How accurately can such predictions be made? The stronger the correlation between the variables in question, the more accurate the predictions. Correlations can range from 0 to –1.00 or +1.00; the greater the departure from 0, the stronger the correlation. Positive numbers mean that as one variable increases, the other increases too. Negative numbers indicate that as one variable increases, the other decreases. For instance, there is a negative correlation between age and the amount of hair on the heads of males: the older they are, the less hair they have.

These basic facts underlie an important method of research sometimes used by social psychologists: the **correlational method**. In this approach, social psychologists attempt to determine whether, and to what extent, different variables are related to each other. This involves carefully measuring each variable, and then performing appropriate statistical tests to determine whether and to what degree the variables are correlated. Perhaps a concrete example will help.

Imagine that a social psychologist wants to find out whether the information posted by users on Facebook is accurate—whether it portrays the users realistically, or presents them as they would like to be (an idealized self-image). Furthermore, imagine that on the basis of previous studies, the researcher hypothesizes that the information people post on Facebook is indeed relatively accurate. How could this idea be tested? One very basic approach, using the correlational method of research, is as follows. First, posters on Facebook would complete measures of their personality (e.g., these could include extraversion, conscientiousness, openness to experience—ones found to be very basic in past research). Then, raters would read the profiles on Facebook and from this information, rate the posters on the same personality dimensions. As a cross-check, other people who know the posters well could also rate them on the same personality dimensions. Next, these sets of information would be compared (i.e., correlated) to see how closely they align. The higher the correlation between these ratings—the ones provided by the posters themselves and people who know them very well (i.e., self and other personality ratings)—the more accurately users of Facebook present themselves. Why? Because the ratings posted by people on Facebook agree with those provided by others who know them personally. In addition, to test the alternative idea that posters try to present themselves in an idealized way, these individuals could be asked to describe their "ideal selves," and this information, too, could be correlated with ratings of their Facebook postings. These basic methods were actually used by Back et al. (2010) in a study designed to find out whether, and to what extent, Facebook postings are accurate with respect to posters' personality. Results offered clear support for the **hypothesis** that these profiles are indeed accurate: Posted profiles closely matched the posters' actual personalities, as measured by personality scales they themselves completed and ratings by friends and family members. In addition, there was little evidence for attempts at idealized self-presentation. On the basis of this research, we can tentatively conclude that Facebook information is accurate and informative about posters' personalities; their personality scores predict their postings, and their postings predict their personality scores. But please emphasize the word *tentatively*, for two important reasons.

correlational method
A method of research in which a scientist systematically observes two or more variables to determine whether changes in one are accompanied by changes in the other.

hypothesis
An as yet unverified prediction concerning some aspect of social behavior or social thought.

First, the fact that two variables are correlated in no way guarantees that they are causally related—that changes in one *cause* changes in the other. On the contrary, the relationship between them may be due to the fact that both variables are related to a third variable, and not really to each other. For instance, in this case, it is possible that people who post on Facebook are simply good at self-presentation—presenting themselves to

others so as to "look good." To the extent that's true, then the correlation between their postings on Facebook and scores on personality tests could reflect this variable. Since they are high in self-presentation skills, their postings and their answers to personality tests both tend to put them in a good light. But in fact the two measures are unrelated to each in any direct or causal way.

Second, it is also possible that posting on Facebook leads to changes in posters' personalities, in the direction of becoming more like the information on Facebook. That may sound a little far-fetched, but it is still possible, and correlational research cannot definitely rule out such possibilities: it can't establish the direction of relationships between variables, just their existence and strength.

Despite these major drawbacks, the correlational method of research is sometimes very useful to social psychologists. It can be used in natural settings where experiments might be very difficult to conduct, and it is often highly efficient: a large amount of information can be obtained in a relatively short period of time. However, the fact that it is generally not conclusive with respect to cause-and-effect relationships is a serious one that leads social psychologists to prefer another method in many instances. It is to this approach that we turn next.

The Experimental Method: Knowledge Through Systematic Intervention

As we have just seen, the correlational method of research is very useful from the point of view of one important goal of science: making accurate predictions. It is less useful, though, from the point of view of attaining another important goal: *explanation*. This is sometimes known as the "why" question because scientists do not merely wish to describe the world and relationships between variables in it: they want to be able to explain these relationships, too.

In order to attain the goal of explanation, social psychologists employ a method of research known as **experimentation** or the **experimental method**. As the heading of this section suggests, experimentation involves the following strategy: One variable is changed systematically, and the effects of these changes on one or more other variables are carefully measured. If systematic changes in one variable produce changes in another variable (and if two additional conditions we describe below are also met), it is possible to conclude with reasonable certainty that there is indeed a causal relationship between these variables: that changes in one do indeed cause changes in the other. Because the experimental method is so valuable in answering this kind of question, it is frequently the method of choice in social psychology. But please bear in mind that there is no single "best" method of research. Rather, social psychologists, like all other scientists, choose the method that is most appropriate for studying a particular topic.

EXPERIMENTATION: ITS BASIC NATURE In its most basic form, the experimental method involves two key steps: (1) the presence or strength of some variable believed to affect an aspect of social behavior or thought is systematically changed and (2) the effects of such changes (if any) are carefully measured. The factor systematically varied by the researcher is termed the **independent variable**, while the aspect of behavior studied is termed the **dependent variable**. In a simple experiment, then, different groups of participants are randomly assigned to be exposed to contrasting levels of the independent variable (such as low, moderate, and high). The researcher then carefully measures their behavior to determine whether it does in fact vary with these changes in the independent variable. If it does—and if two other conditions are also met—the researcher can tentatively conclude that the independent variable does indeed cause changes in the aspect of behavior being studied.

To illustrate the basic nature of experimentation in social psychology, we'll use the following example. Suppose that a social psychologist is interested in the question,

experimentation (experimental method)
A method of research in which one or more factors (the independent variables) are systematically changed to determine whether such variations affect one or more other factors (dependent variables).

independent variable
The variable that is systematically changed (i.e., varied) in an experiment.

dependent variable
The variable that is measured in an experiment.

Does exposure to violent video games increase the likelihood that people will aggress against others in various ways (e.g., verbally, physically, spreading false rumors, or posting embarrassing photos of them on the Internet; see Figure 1.12). How can this possibility be investigated by using the experimental method? Here is one possibility.

Participants in the experiment could be asked to play a violent or nonviolent video game. After these experiences in the research, they would be placed in a situation where they could, if they wished, aggress against another person. For instance, they could be told that the next part of the study is concerned with taste sensitivity and asked to add as much hot sauce as they wish to a glass of water that another person will drink. Participants would taste a sample in which only one drop of sauce has been placed in the glass, so they would know how hot the drink would be if they added more than one drop. Lots of sauce would make the drink so hot that it would truly hurt the person who consumed it.

If playing aggressive video games increases aggression against others, then participants who played such games would use more hot sauce—and so inflict more pain on another person—than participants who examined the puzzle. If results indicate that this is the case, then the researcher could conclude, at least tentatively, that playing aggressive video games does increase subsequent, overt aggression. The researcher can offer this conclusion because if the study was done correctly, the only difference between the experiences of the two groups during the study is that one played violent games and the other did not. As a result, any difference in their behavior (in their aggression) can be attributed to this factor. It is important to note that in experimentation, the independent variable—in this case, exposure to one or another type of video game—is systematically changed by the researcher. In the correlational method, in contrast, variables are not altered in this manner; rather, naturally occurring changes in them are simply observed and recorded. By the way, research findings reported over several



FIGURE 1.12 The Experimental Method: Using It to Study the Effects of Violent Video Games

Does playing violent video games such as the one shown here increase the tendency to aggress against others? Using the experimental method, social psychologists can gather data on this important issue—and in fact, have already done so!

decades do indicate that regular exposure to violence in the media or in video games does seem to increase aggression against others, and that this link is in fact a casual one: regular or frequent exposure to violent content reduces sensitivity to such materials, and enhances aggressive thoughts and emotions (e.g., Krahe, Moller, Huesmann, Kirwill, Felber, & Berger, 2011).

EXPERIMENTATION: TWO KEY REQUIREMENTS FOR ITS SUCCESS Earlier, we referred to two conditions that must be met before a researcher can conclude that changes in an independent variable have caused changes in a dependent variable. Let's consider these conditions now. The first involves what is termed **random assignment of participants to experimental conditions**. This means that all participants in an experiment must have an equal chance of being exposed to each level of the independent variable. The reason for this rule is simple: If participants are not randomly assigned to each condition, it may later be impossible to determine if differences in their behavior stem from differences they brought with them to the study, from the impact of the independent variable, or both. For instance, imagine that in the study on video games, all the people assigned to the violent game come from a judo club—they practice martial arts regularly—while all those assigned to play the other game come from a singing club. If those who play the violent games show higher levels of aggression, what does this tell us? Not much! The difference between the two groups stem from the fact that individuals who already show strong tendencies toward aggression (they are taking a judo class) are more aggressive than those who prefer singing; playing violent video games during the study might be completely unrelated to this difference, which existed prior to the experiment. As result, we can't tell why any differences between them occurred; we have violated random assignment of people to experimental treatments, and that makes the results virtually meaningless.

The second condition essential for successful experimentation is as follows: Insofar as possible, all factors other than the independent variable that might also affect participants' behavior must be held constant. To see why this is so, consider what will happen if, in the study on video games, two assistants collect the data. One is kind and friendly, the other is rude and nasty. By bad luck, the rude assistant collects most of the data for the aggressive game condition and the polite one collects most of the data from the nonaggressive game condition. Again, suppose that participants in the first group are more aggressive toward another person. What do the findings tell us? Again, virtually nothing, because we can't tell whether it was playing the aggressive video game or the rude treatment they received from the assistant that produced higher aggression. In situations like this, the independent variable is said to be *confounded* with another variable—one that is not under systematic investigation in the study. When such confounding occurs, the findings of an experiment may be largely uninterpretable (see Figure 1.13).

In sum, experimentation is, in several respects, the most powerful of social psychology's methods. It certainly isn't perfect—for example, since it is often conducted in laboratory settings that are quite different from the locations in which social behavior actually occurs, the question of *external validity* often arises: To what extent can the findings of experiments be generalized to real-life social situations and perhaps people different from those who participated in the research? And there are situations where, because of ethical or legal considerations, it can't be used. For instance, it would clearly be unethical to expose couples to conditions designed to weaken their trust in one another, or to expose research participants to a kind of television programming that may cause them to harm themselves. But in situations where it is appropriate and is used with skill and care, however, the experimental method can yield results that help us to answer complex questions about social behavior and social thought. Overall, though, please keep the following basic point in mind: there is no single best method of conducting research in social psychology. Rather, all methods offer advantages and disadvantages, so the guiding principle is that the method that is most appropriate to answering the questions being investigated is the one that should be used.

random assignment of participants to experimental conditions
A basic requirement for conducting valid experiments. According to this principle, research participants must have an equal chance of being exposed to each level of the independent variable.

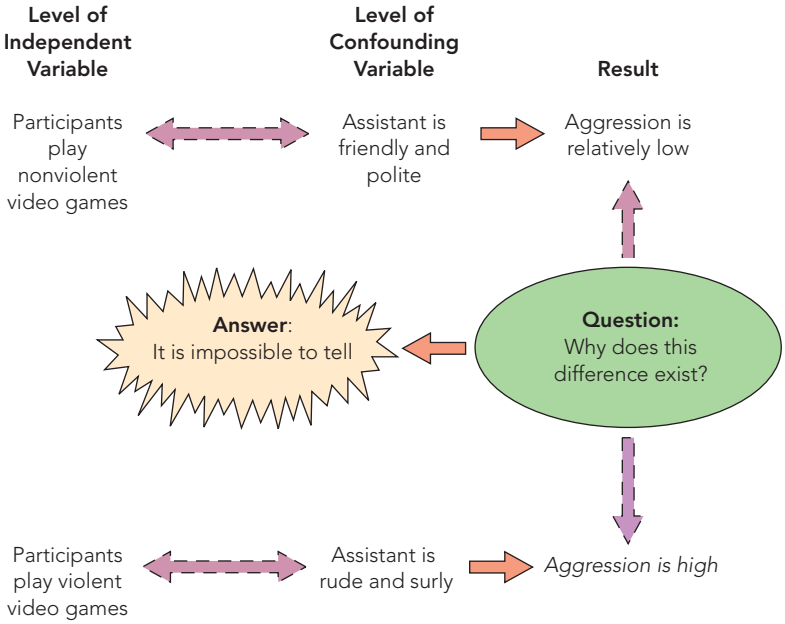


FIGURE 1.13 Confounding of Variables: A Fatal Flaw in Experimentation

In a hypothetical experiment designed to investigate the effects of playing violent video games on aggression, the independent variable is confounded with another variable, the behavior of the assistants conducting the study. One assistant is kind and polite and the other is rude and surly. The friendly assistant collects most of the data in nonviolent game condition, while the rude assistant collects most of the data in the violent game condition. Findings indicate that people who play the violent video games are more aggressive. But because of confounding of variables, we can't tell whether this is a result of playing these games or the assistant's rude treatment. The two variables are confounded, and the experiment doesn't provide useful information on the issue it is designed to study.

Further Thoughts on Causality: The Role of Mediating Variables

Earlier, we noted that social psychologists often use experimentation because it is helpful in answering questions about causality: Do changes in one variable produce (cause) changes in another? That is a very valuable kind of information to have because it helps us understand what events, thoughts, or situations lead to various outcomes—more or less helping, more or less aggression, more or less prejudice. Often, though, social psychologists take experimentation one step further in their efforts to answer the question of why—to understand why one variable produces changes in another. For instance, returning to the video game study described above, it is reasonable to ask, Why does playing such games increase aggression? Because it induces increased thoughts about harming others? Reminds people of real or imagined wrongs they have suffered at the hands of other people? Convinces them that aggression is okay since it leads to high scores in the game?

To get at this question of underlying processes, social psychologists often conduct studies in which they measure not just a single dependent variable, but other factors that they believe to be at work—factors that are influenced by the independent variable and then, in turn, affect the dependent measures. For instance, in this study, we could measure participants' thoughts about harming others and their beliefs about when and whether aggression is acceptable social behavior to see if these factors help explain why playing violent video games increases subsequent aggression. If they do, then they are termed **mediating variables**, ones that intervene between an independent variable (here, playing certain kinds of video games) and changes in social behavior or thought.

mediating variable
A variable that is affected by an independent variable and then influences a dependent variable. Mediating variables help explain why or how specific variables influence social behavior or thought in certain ways.

The Role of Theory in Social Psychology

There is one more aspect of social psychological research we should consider before concluding. As we noted earlier, in their research, social psychologists seek to do more than simply describe the world: they want to be able to *explain* it too. For instance, social psychologists are not interested in merely stating that racial prejudice is common in the United States (although, perhaps, decreasing); they want to be able to explain why some people are more prejudiced toward a particular group than are others. In social psychology, as in all branches of science, explanation involves the construction of theories—frameworks for explaining various events or processes. The procedure involved in building a theory goes something like this:

1. On the basis of existing evidence, a theory that reflects this evidence is proposed.
2. This theory, which consists of basic concepts and statements about how these concepts are related, helps to organize existing information and makes predictions about

observable events. For instance, the theory might predict the conditions under which individuals acquire racial prejudice.

3. These predictions, known as *hypotheses*, are then tested by actual research.
4. If results are consistent with the theory, confidence in its accuracy is increased. If they are not, the theory is modified and further tests are conducted.
5. Ultimately, the theory is either accepted as accurate or rejected as inaccurate. Even if it is accepted as accurate, however, the theory remains open to further refinement as improved methods of research are developed and additional evidence relevant to the theory's predictions is obtained.

This may sound a bit abstract, so let's turn to a concrete example. Suppose that a social psychologist formulates the following theory: When people believe that they hold a view that is in the minority, they will be slower to state it and this stems not from the strength of their views, but from reluctance to state minority opinions publicly where others will hear and perhaps disapprove of them for holding those views. This theory would lead to specific predictions—for instance, the minority slowness effect will be reduced if people can state their opinions privately (e.g., Bassili, 2003). If research findings are consistent with this prediction and with others derived from the theory, confidence in the theory is increased. If findings are not consistent with the theory, it will be modified or perhaps rejected, as noted above.

This process of formulating a theory, testing it, modifying the theory, testing it again, and so on lies close to the core of the scientific method, so it is an important aspect of social psychological research (see Figure 1.14). Thus, many different theories relating to important aspects of social behavior and social thought are presented in this book.

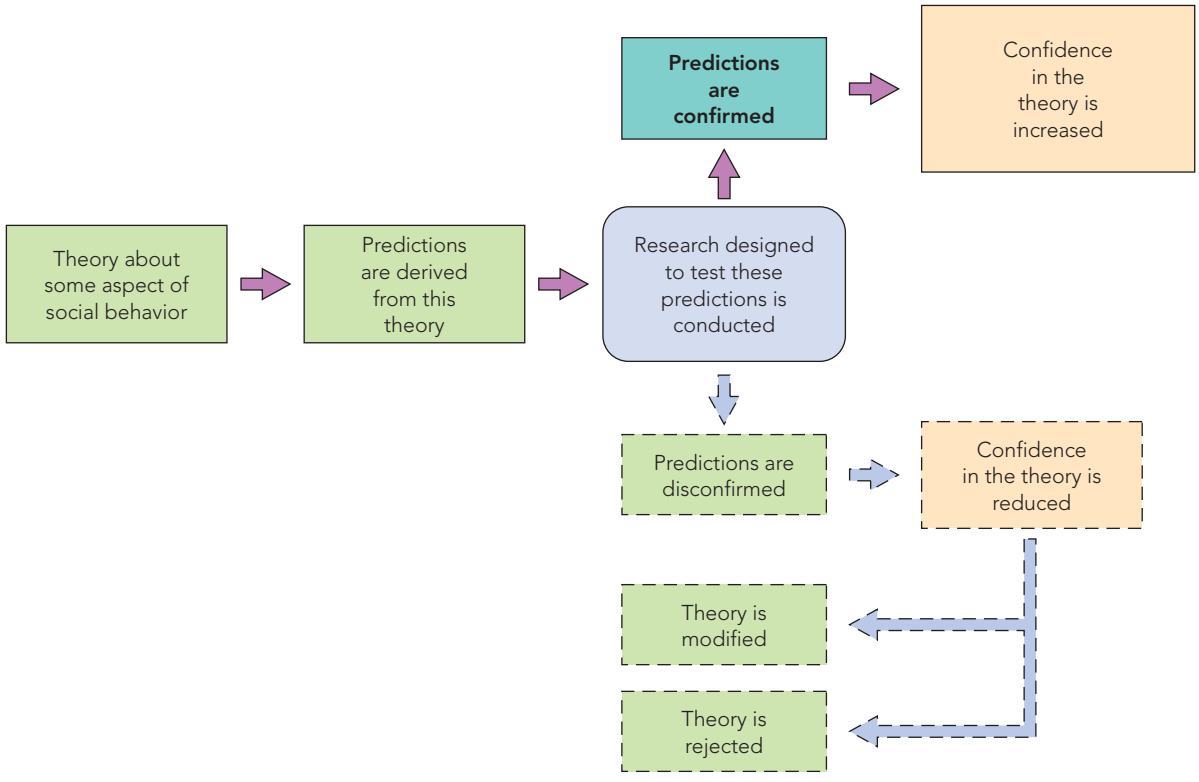


FIGURE 1.14 The Role of Theory in Social Psychological Research

Theories both organize existing knowledge and make predictions about how various events or processes will occur. Once a theory is formulated, hypotheses derived logically from it are tested through careful research. If results agree with the predictions, confidence in the theory is increased. If results disagree with such predictions, the theory may be modified or ultimately rejected as false.

Two final points. First, theories are never proven in any final, ultimate sense; rather, they are always open to test, and are accepted with more or less confidence depending on the weight of available evidence. Second, research is not undertaken to prove or verify a theory; it is performed to gather evidence relevant to the theory. If a researcher sets out to "prove" her or his pet theory, this is a serious violation of the principles of scientific skepticism, objectivity, and open-mindedness described on page 06.



KEY POINTS

- With **systematic observation,** behavior is carefully observed and recorded. In naturalistic observation, such observations are made in settings where the behavior naturally occurs.
- **Survey methods** often involve large numbers of people who are asked to respond to questions about their attitudes or behavior.
- When the **correlational method** of research is employed, two or more variables are measured to determine how they might be related to one another.
- The existence of even strong correlations between variables does not indicate that they are causally related to each other.
- **Experimentation** involves systematically altering one or more variables **(independent variables)** in order to determine whether changes in this variable affect some aspect of behavior **(dependent variables)**.
- Successful use of the **experimental method** requires **random assignment of participants to conditions** and holding all other factors that might also influence behavior constant so as to avoid confounding of variables.
- Although it is a very powerful research tool, the experimental method is not perfect—questions concerning the external validity of findings so obtained often arise. Furthermore, it cannot be used in some situations because of practical or ethical considerations.
- Research designed to investigate **mediating variables** adds to understanding of how specific variables influence certain aspects of social behavior or social thought.
- Theories are frameworks for explaining various events or processes. They play a key role in social psychological research.

The Quest for Knowledge and the Rights of Individuals: In Search of an Appropriate Balance

In their use of experimentation, correlation, and systematic observation, social psychologists do not differ from researchers in other fields. One technique, however, does seem to be unique to research in social psychology: **deception**. This technique involves efforts by researchers to withhold or conceal information about the purposes of a study from participants. The reason for doing so is simple: Many social psychologists believe that if participants know the true purposes of a study, their behavior in it will be changed by that knowledge. Thus, the research will not yield valid information about social behavior or social thought, unless deception is employed.

deception
A technique whereby researchers withhold information about the purposes or procedures of a study from people participating in it.

Some kinds of research do seem to require the use of temporary deception. For example, consider the video game study described above. If participants know that the purpose of a study is to investigate the impact of such games, isn't it likely that they might lean over backward to avoid showing it? Similarly, consider a study of the effects of physical appearance on attraction between strangers. Again, if participants know that the

researcher is interested in this topic, they might work hard to avoid being influenced by a stranger's appearance. In this and many other cases, social psychologists feel compelled to employ temporary deception in their research (Suls & Rosnow, 1988). However, the use of deception raises important ethical issues that cannot be ignored.

First, there is the chance, however slim, that deception may result in some kind of harm to the people exposed to it. They may be upset by the procedures used or by their own reactions to them. For example, in several studies concerned with helping in emergencies, participants were exposed to seemingly real emergency situations. For instance, they overheard what seemed to be a medical emergency—another person having an apparent seizure (e.g., Darley & Latané, 1968). Many participants were strongly upset by these staged events, and others were disturbed by the fact that although they recognized the need to help, they failed to do so. Clearly, the fact that participants experienced emotional upset raises complex ethical issues about just how far researchers can go when studying even very important topics such as this one.

We should hasten to emphasize that such research represents an extreme use of deception: generally, deception takes much milder forms. For example, participants may receive a request for help from a stranger who is actually an assistant of the researchers; or they may be informed that most other students in their university hold certain views when in fact they do not. Still, even in such cases, the potential for some kind of harmful effects to participants exists and this is a potentially serious drawback to the use of deception.

Second, there is the possibility that participants will resent being "fooled" during a study and, as a result, they will acquire negative attitudes toward social psychology and psychological research in general; for instance, they may become suspicious about information presented by researchers (Kelman, 1967). To the extent such reactions occur—and recent findings indicate that they do, at least to a degree (Epley & Huff, 1998)—they have disturbing implications for the future of social psychology, which places much emphasis on scientific research.

Because of such possibilities, the use of deception poses something of a dilemma to social psychologists. On the one hand, it seems essential to their research. On the other, its use raises serious problems. How can this issue be resolved? While opinion remains somewhat divided, most social psychologists agree on the following points. First, deception should never be used to persuade people to take part in a study; withholding information about what will happen in an experiment or providing misleading information in order to induce people to take part in it is definitely not acceptable (Sigall, 1997). Second, most social psychologists agree that temporary deception may sometimes be acceptable, provided two basic safeguards are employed. One of these is **informed consent**—giving participants as much information as possible about the procedures to be followed before they make their decision to participate. In short, this is the opposite of withholding information in order to persuade people to participate. The second is careful **debriefing**—providing participants with a full description of the purposes of a study after they have participated in it (see Figure 1.15). Such information should also include an explanation of deception, and why it was necessary to employ it.

Fortunately, existing evidence indicates that together, informed consent and thorough debriefing can substantially reduce the potential dangers of deception (Smith & Richardson, 1985). For example, most participants report that they view temporary deception as acceptable, provided that potential benefits outweigh potential costs and if there is no other means of obtaining the information sought (Rogers, 1980; Sharpe, Adair, & Roese, 1992). However, as we noted above, there is some indication that they do become somewhat more suspicious about what researchers tell them during an experiment; even worse, such increased suspiciousness seems to last over several months (Epley & Huff, 1998).

Overall, then, it appears that most research participants do not react negatively to temporary deception as long as its purpose and necessity are subsequently made clear. However, these findings do not mean that the safety or appropriateness of deception

informed consent
A procedure in which research participants are provided with as much information as possible about a research project before deciding whether to participate in it.

debriefing
Procedures at the conclusion of a research session in which participants are given full information about the nature of the research and the hypothesis or hypotheses under investigation.



FIGURE 1.15 Careful Debriefing: A Requirement in Studies Using Deception
After an experimental session is completed, participants should be provided with thorough debriefing—full information about the experiment's goals and the reasons why temporary deception is considered necessary.

should be taken for granted (Rubin, 1985). On the contrary, the guiding principles for all researchers planning to use this procedure should be: (1) Use deception only when it is absolutely essential to do so—when no other means for conducting the research exists; (2) always proceed with caution; and (3) make certain that every possible precaution is taken to protect the rights, safety, and well-being of research participants. In terms of the latter, all universities in the United States who receive federal funding must have an Institutional Review Board to review the ethics, including a cost–benefit analysis when deception is to be employed, for all proposed research involving human participants.

KEY POINTS

- **Deception** involves efforts by social psychologists to withhold or conceal information about the purposes of a study from participants.
- Most social psychologists believe that temporary deception is often necessary in order to obtain valid research results.
- However, most social psychologists view deception as acceptable only when important safeguards are employed: **informed consent** and thorough **debriefing**.

Getting the Most Out of This Book: A User's Guide

A textbook that is hard to read or understand is like a dull tool: it really can't do what it is designed to do. We are fully aware of this fact, so we have tried our best to make this book as easy to read as possible, and have included a number of features designed to make it more enjoyable—and useful—for you. Here is a brief overview of the steps we've taken to make reading this book a pleasant and informative experience.

First, each chapter begins with an outline of the topics to be covered. This is followed by a chapter-opening story that "sets the stage," and explains how the topics to be covered are related to important aspects of our everyday lives. Within each chapter, key terms are printed in **boldface** type and are followed by a definition. These terms are also defined in the margins of the pages on which they are first mentioned, as well as in a glossary at the end of the book. To help you understand what you have read, each major section is followed by a list of Key Points—a brief summary of the major points. All figures and tables are clear and simple, and most contain special labels and notes designed to help you understand them (see Figure 1.8 for an example). Finally, each chapter ends with a Summary and Review. Reviewing this section can be an important aid to your studying.

Second, this book has an underlying theme, which we have already stated (see page 6), but want to emphasize again: Social psychology seeks basic principles concerning social thought and social behavior—principles that apply very generally, in all cultures and settings. But it recognizes that the context in which the social side of life occurs is very important. Because of the growing role of technology in our lives, the ways in which we interact with other people have changed and now often occur via cell phones, computers, and other electronic devices rather than in face-to-face encounters. We believe that the basic principles of social psychology apply to these new contexts too, but that their accuracy in the "cyber" or "electronic" world must be established by careful research. To take account of this major change in the settings and modes of expression of social behavior, we report research concerning social networks, the Internet, and related topics throughout the book. In addition, to call special attention to their growing importance, we include special sections with up-to-date research in each chapter, titled "Social Life in a Connected World." A few examples: Dating on the Internet; Humiliating Others Through the Web; Helping in Social Networks. We think that these sections will take account of important societal changes that are, indeed, strongly affecting the nature and form of the social side of life.

An additional theme in modern social psychology—and one we have already described—is growing interest in the role of emotion in our social thought and actions. To highlight recent advances in our knowledge of this topic, we will include another type of special section titled "Emotions and…" (for example, "Emotions and Attitudes," "Emotions and Aggression," "Emotions and Group Life"). These sections illustrate the powerful influence of the feeling side of social life, and are based on current and informative research in the field.

We think that together, these features will help you get the most out of this book, and from your first contact with social psychology. Good luck! And may your first encounter with our field prove to be a rich, informative, valuable, and enjoyable experience.



SUMMARY *and* REVIEW

- Social psychology is the scientific field that seeks to understand the nature and causes of individual behavior and thought in social situations. It is scientific in nature because it adopts the values and methods used in other fields of science. Social psychologists adopt the scientific method because "common sense" provides an unreliable guide to social behavior, and because our personal thought is influenced by many potential sources of bias. Social psychology focuses on the behavior of individuals, and seeks to understand the causes of social behavior and thought, which can involve the behavior and appearance of others, social cognition, environmental factors, cultural values, and even biological and genetic factors. Social psychology seeks to establish basic principles of social life that are accurate across huge cultural differences and despite rapid and major changes in social life.

- Important causes of social behavior and thought include the behavior and characteristics of other people, cognitive processes, emotion, culture, and genetic factors. Social psychologists currently recognize that social thought and social behavior are two sides of the same coin, and that there is a continuous, complex interplay between them. There is growing interest among social psychologists in the role of emotion in social thought and social behavior. The formation and development of **relationships** is another major trend in the field. Yet another major trend involves growing interest in social neuroscience—efforts to relate activity in the brain to key aspects of social thought and behavior.

- Our behavior and thought is often shaped by factors of which we are unaware. Growing attention to such implicit (nonconscious) processes is another major theme of modern social psychology. Social psychology currently adopts a **multicultural perspective.** This perspective recognizes the importance of cultural factors in social behavior and social thought, and notes that research findings obtained in one culture do not necessarily generalize to other cultures. With **systematic observation,** behavior is carefully observed and recorded. In naturalistic observation, such observations are made in settings where the behavior naturally occurs. **Survey methods** often involve large numbers of people who are asked to